

Project:

Interactive Data Visualization Dashboard for TB Using D3

Deadline: 30th November 2024

Project Objective

To create an interactive data visualization dashboard that analyzes relationships between entities concerning various dimensions such as location, timestamp, and type of relationship. The dashboard will employ diverse visualizations to present data in an insightful and user-friendly manner, enhancing the ability to discern patterns and insights within the dataset.

Dataset Links

- [TB Portals Data Sharing](#)
- [TB-DIAH Data Hub](#)

Data Requirements

- **Dataset:** Utilize the TB dataset, focusing on attributes such as:
 - **Relationship Type:** Type of relationship between entities (e.g., collaboration, conflict).
 - **Location:** Geographic coordinates or identifiers of entities.
 - **Timestamp:** Date and time indicating when the relationship was recorded.
 - **Entity IDs:** Unique identifiers for each entity involved in relationships.

- **Path:** Descriptive paths indicating the nature of relationships or interactions.
 - **Data Preparation:** Preprocess the dataset to ensure it is clean, structured appropriately for visualizations, and ready for hierarchical analysis if applicable.
-

Key Visualizations and Features

1. Force-Directed Graph for Relationship Mapping

- **Description:** A network visualization that displays entities as nodes and their relationships as edges.
- **Features:**
 - **Nodes** represent entities, while **edges** signify relationships between them.
 - **Color Coding:** Different colors represent various relationship types for quick identification.
 - **Interactivity:**
 - Hovering over nodes reveals detailed information.
 - Click actions to expand nodes for deeper insights.
 - Zooming and panning to explore the network.
 - **Filters:** Dynamic filters allow users to select specific relationship types, enhancing data exploration.

2. Map Chart for Geographic Representation

- **Description:** Visualizes entities based on their geographic locations.
- **Features:**
 - **Pins** on the map indicate entity locations, color-coded by relationship type.

- **Interactive Features:**
 - Zoom and pan functionality to explore different geographic areas.
 - Hovering over pins displays details about specific locations, while clicking highlights related entities.
- **Filters:** Users can filter by region or relationship type to focus on specific data points.

3. Timeline Visualization with Animation

- **Description:** An animated timeline illustrating the evolution of relationships over time.
- **Features:**
 - Data points are plotted against timestamps, showing changes and developments in relationships.
 - A playable animation allows users to observe how relationships progress over time.
 - **Interactivity:** Users can pause the animation to examine specific timeframes.
 - **Filters:** A date range slider enables users to filter visible data based on selected time intervals.
- **Example:**

<https://observablehq.com/@mbostock/the-wealth-health-of-nations>

4. Hierarchical Tree Map for Entity Categorization

- **Description:** Represents relationships in a hierarchical format based on entity types or categories.
- **Features:**

- **Nested Rectangles** illustrate different levels of the hierarchy (e.g., departments within organizations).
- Size and color reflect metrics such as the frequency of relationships or intensity.
- **Interactivity:** Drill-down functionality allows exploration of deeper hierarchical levels, along with breadcrumb navigation for easy backtracking.
- **Filters:** Hierarchical level filters provide tailored views to investigate specific categories.

5. Sunburst Chart for Hierarchical Data (Bonus Visualization)

- **Description:** A radial chart depicting hierarchical relationships in a circular layout.
- **Features:**
 - Inner rings represent higher hierarchy levels; outer rings show subcategories.
 - Color-coded by relationship type or category for quick visual distinction.
 - **Interactivity:** Hovering reveals details; clicking allows users to zoom into specific categories.
 - **Filters:** Synchronization with other visualizations based on user selection enhances the exploration experience.

Advanced Interactivity Features

- **Focus-Plus-Context (Zoom and Detail):** Users can zoom into specific visualization areas while maintaining an overview of the broader context.

- **Hover and Tooltip Details:** Enhanced tooltips provide in-depth information, including entity details, relationship strength, timestamps, and other metrics.
 - **On-Click Actions:** Clicking on nodes or visual elements triggers cross-visualization interactions, allowing users to focus on related data.
 - **Dynamic Filtering and Sorting:** Multiple filters for relationship types, locations, timestamps, and metrics, along with sorting options to streamline data analysis.
 - **Interactive Legend:** A dynamic legend allows toggling the visibility of relationship types across all visualizations in real time.
-

Bonus Features for Additional Points

1. **Exceptional Interactivity** (5% bonus): Implement advanced focus-plus-context features and intuitive custom tooltips.
2. **Animated Transitions and Smooth Zooming** (5% bonus): Ensure high-quality animations for zooming, expanding, and collapsing elements.
3. **Responsive Design and Compatibility** (5% bonus): Optimize the dashboard for compatibility across devices and browsers.
4. **Advanced Filtering and Sorting Options** (5% bonus): Offer comprehensive filtering and sorting mechanisms for refined data exploration.
5. **Customizable and Exportable Views** (5% bonus): Enable users to customize visualization settings and download images in various formats.
6. **Enhanced Tooltip Information and Insights** (5% bonus): Tooltips containing detailed, dynamic insights based on entity relationships.
7. **Dynamic and Interactive Legend** (5% bonus): A responsive legend that updates visualizations in real time as categories are toggled.
8. **Dashboard State Persistence** (5% bonus): Save user settings such as filters and zoom levels in local storage.

9. **Hierarchical Data Drill-Down with Breadcrumb Navigation** (5% bonus): Smooth navigation through hierarchical data with breadcrumb support.
10. **Cross-Visualization Interaction** (5% bonus): Enable selections in one visualization to update related data across others in real time.
-

Project Deliverables

1. Complete Dashboard with All Visualizations

- Developed using D3.js, SVG, CSS, and JavaScript.
- Each visualization incorporates specified interactivity, filtering, and focus-plus-context features.

2. Documentation and Presentation

- Comprehensive documentation outlining design choices, data preprocessing steps, and interactive features.
 - A short video demo or live presentation illustrating key functionalities and insights.
-

Grading and Evaluation

Component	Marks
Core Visualizations and Features	150
Advanced Interactivity Features	50
Bonus Features	50
Total	250

Breakdown of Core Visualizations and Features (150 Marks)

Feature	Marks
Force-Directed Graph	30
Map Chart	30
Timeline Visualization with Animation	30
Hierarchical Tree Map	30
Sunburst Chart	30
Subtotal	150

Breakdown of Advanced Interactivity Features (50 Marks)

Feature	Marks
Focus-Plus-Context (Zoom and Detail)	10
Hover and Tooltip Details	10
On-Click Actions	10
Dynamic Filtering and Sorting	10
Interactive Legend	10
Subtotal	50

Breakdown of Bonus Features (50 Marks)

Feature	Marks
Exceptional Interactivity	5
Animated Transitions and Smooth Zooming	5
Responsive Design and Compatibility	5
Advanced Filtering and Sorting Options	5
Customizable and Exportable Views	5
Enhanced Tooltip Information and Insights	5
Dynamic and Interactive Legend	5
Dashboard State Persistence	5
Hierarchical Data Drill-Down with Breadcrumb Navigation	5
Cross-Visualization Interaction	5
Subtotal	50